Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/23/2006
PI Name:	Paloski, William H Ph.D.		
Project Title:	Spatial Reorientation of Sensorimotor Balance Control in Altered Gravity		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77204-6015	**Congressional District:**	18
Comments:	PI moved to University of Houston in September 2008, from NASA JSC.		
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	10/01/1999	**End Date:**	02/01/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:	Shuttle		
Flight Assignment:	Pre-/Postflight Only In flight development phase (data collection has begun)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Wood, Scott Ph.D. (Universities Space Research Association) Kaufman, Galen Ph.D. (University of Texas Medical Branch) Reschke, Millard Ph.D. (NASA Johnson Space Center) Black, Franklin M.D. (Legacy Health Sysytem)		
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Disruption of balance control is one of the most significant post-flight manifestations of sensorimotor adaptation to microgravity. This investigation examines changes in spatial processing of sensorimotor function following adaptation to microgravity, and explores the feasibility of forcing changes between motor program sets optimized for different gravitoinertial environments. This study includes pre- and post-flight measurements obtained on ten astronaut subjects (preferably first-time fliers) following short-duration Shuttle missions. Each astronaut participates in one familiarization training session (including standard posturography testing), three pre-flight data sessions, and five post-flight data sessions on R+0, 2, 3, 4, and 8 days. Each test session includes posture measurements, and three sessions (L-60, L-30 and R+3) include additional posture measurements immediately after exposure to short-radius centrifugation. In addition, testing on normative subjects support data interpretation and establish ground-based controls. Our first specific aim is to examine adaptive changes in the spatial reference frame used for coding orientation and motion as a function of space flight. By examining the effects of head tilt on balance control, we will test our hypothesis that there is a reorientation of central vestibular processing from a gravitational frame of reference to a head frame of reference as a function of adaptation to microgravity. Our second specific aim is to examine the feasibility of altering the re-adaptation process following space flight by providing discordant canal-otolith-somatosensory stimuli using short-radius pitch centrifugation. This study specifically addresses Neurovestibular Adaptation questions of the Critical Path Roadmap (CPR Risks 15 and 16) related to the physiological basis for disruption of balance control and spatial disorientation following g-transitions. Additional post-flight posturography measures obtained during this study will expand our existing database and increase our ability to predict recovery curves from mathematical models for individuals based on pre-flight postural performance. The incorporation of head tilts during posturography addresses our general hypothesis for the physiological basis for the disruption of balance control following g-transitions, and provides an enhanced technique to determine crew readiness to return to normal activities following return to Earth. The incorporation of short-radius centrifugation during the post-flight period addresses the ability of crewmembers to adapt to multiple gravitoinertial environments and switch between them, e.g., as required for intermittent exposure to artificial gravity. The results of this study will also assess the potential risk for disrupting the re-adaptation process by exposure to discordant sensory stimuli during low levels of centrifugation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research can be used by the NASA research and operational community to evaluate the feasibility of using short arm centrifugation as an in-flight sensorimotor countermeasure. The results are of use to clinical and military researchers interested in adaptive changes in spatial processing to altered gravito-inertial environments, and the fragility of balance control adaptive processes. Findings of the ground-based study on static and dynamic head movements during computerized dynamic posturography may help to improve the sensitivity of the test paradigm in selected clinical populations.
Task Progress:	Normative studies: A manuscript describing the results from one of our normative studies was published in Gait and Posture (Paloski et al, 2006). This study was useful to optimize the astronaut posture protocol by: restricting head tilts to the pitch plane, limiting the frequency of the dynamic tilts to 0.33 Hz, reducing the amplitude of the tilts to 20° and performing statistical power analyses to determine the number of trials required for each condition. A second manuscript is in revision describing the results of a companion eccentric rotation study that was used to finalize the centrifuge protocol for Specific Aim 2 (Wood et al., 2006). Pre- and Post-flight Measures – Specific Aim 1: Our first specific aim is to examine adaptive changes in the spatial reference frame used for coding orientation and motion as a function of space flight. We predicted the use of a head spatial reference frame will be characterized by differences in the landing day measurements relative to those observed before flight. These include decreased postural stability as evidenced by increased sway amplitudes and decreased equilibrium scores, with greater differences between pre- and post-flight during head tilt conditions. Nine astronauts have completed this study to date. In this past reporting year, pre- and post-flight data was obtained on four additional participants from the STS-121 and STS-115 missions. We also resumed an ongoing normative study designed to obtain repeated postural measures in healthy age, gender and height-matched control subjects using the same timeline scheduling as their astronaut counterparts. Pre-flight scores for head erect, forward, backward, and moving at 0.33 Hz in the pitch plane were all in the normal range of the performance distributions. Consistent with our previous post-flight studies, performance on landing day was substantially decreased for these conditions with absent visual and altered proprioceptive feedback. The most striking result on landing day; however, was on trials requiring active head movements at 0.33 Hz. On landing day, all nine subjects fell on at least one SOT-5 trial with active head movements (falls on 14 of 18 trials overall). The recovery curves for these crewmembers for trials with head erect followed similar trajectories to previous astronaut data recorded by our laboratory. The recovery profile for trials with the head movements, in contrast, had a similar recovery time constant but at lower performance levels compared with the head erect trials. Pre- and Postflight Measures – Specific Aim 2: Our second specific aim is to examine the feasibility of altering the re-adaptation process following space flight by providing discordant canal-otolith-somatosensory stimuli using short-radius pitch axis centrifugation. The centrifuge profile utilized in this study represents a composite of different profiles that appeared to disrupt the recovery of one Spacelab astronaut as late as R+5 as described above. Although there were some interesting modulations of vertical and horizontal vergence eye movements during centrifugation, there were no apparent differences in these responses between pre- and post-flight. There were also no consistent differences between motion perception reports from pre- to post-flight. Of the 9 participants to date, the centrifugation did not substantially disrupt postural performance either pre- or post-flight.
Bibliography Type:	Description: (Last Updated: 02/09/2021)

Articles in Peer-reviewed Journals	Paloski WH, Wood SJ, Feiveson AH, Black FO, Hwang EY, Reschke MF. "Destabilization of human balance control by static and dynamic head tilts." Gait Posture 2006 Apr; 23(3):315-23. PMID:15961313 , Apr-2006